<u>Java Exercise 3</u>

1. Print the sum of first n numbers. If N is 3 then print the sum of 1+2+3 to the user. Get n from the user

2. Print the multiplication table by getting the n from the user.

   For n = 2 the table should be printed like shown below

   2 Times Table

   2 x 0 = 0
   2 x 1 = 2
   2 x 2 = 4
   2 x 3 = 6
   2 x 4 = 8
   2 x 5 = 10
   2 x 6 = 12
   2 x 7 = 14
   2 x 8 = 16
   2 x 9 = 18
   2 x 10 = 20

3. Provide the option of adding two numbers to the user until the user wants to exit. (Use Do While Loop)

4. Read a number from the user and determine **<u>how many digits in the number</u>** are even ,odd and prime. For example ,if the user enters an input number 934 , the number of even digits in the number is 1 (that is 4) , odd digits is 2 (3 and 9) , prime digits is 1 (3)

   The output of your program should be

   The number of even digits is 1
   The number of odd digits is 2
   The number of prime digits is 1

5. Read 5 numbers as input from the user. Display the count of odd and even numbers entered (use For Loop)
6. Read an integer from the user and Display the reverse of the digits of an Integer number to the user. (Use While Loop)

7. A company has decided to give bonus for their employees who have completed three years of service. The amount of bonus is calculated as 500*(No. of years of Experience - 3). Given the details of the employees in the company as shown below and an employee name, write an algorithm and the subsequent Java code to determine the bonus amount that the employee will receive. If the experience of the employee is greater than 3 then print the amount to be received and the mobile number of the employee and print 'Not eligible for bonus' otherwise. For example, if the name of the employee given is 'Rajesh' then print 'Not eligible for bonus. If the employee name given is 'Kumar' then the amount to be received is 7000, therefore print 7000 and his mobile number 9012345621. Read mobile number as an integer.

Name of Employee	Mobile number of student	Experience
Kumar	9012345621	17
Dinesh	8143567890	7
Ganesh	7114567213	13
Rajesh	9098456743	2
Rakesh	8159056784	9

8. A number is said to have the property of Odd Sum if the sum of the number and its reverse is odd. For example, 36 is having the "Odd sum" property since 36 + 63 = 99 which is odd. Given a number 'n', write an algorithm and the subsequent Java program to check whether the given number 'n' has the 'odd sum property'. If the given number has the 'odd sum property', print "Odd Sum", else print " No odd sum".

   Input Format

   A number, n

   Output Format

   Print either "Odd sum" or "No odd sum"

9. Compute the area of an equilateral triangle. The formula is given below

   (S is the length of any side of the triangle)

$$\frac{\sqrt{3}}{4}s^2$$

10. Write a Java Program to read a number n from the user and determine the result of n factorial.
11. Read the names of 10 countries from the user and display only the countries that start with an “U” to the User.
12. Read a string from the user and count the number of vowels in the string and display the count to the user.
13. Read two strings from the user. Check if the two strings are equal**. Do not use any inbuilt functions for string comparison** to complete this code.
14. Create a menu based calculator application. The user should be able to enter two numbers and select an operation – (Add /subtract/multiply/ divide). The result of hte as many times he wishes without exiting the application.
15. Print the following pattern. Get the number of lines to be printed from the user.

```
*
**
***
****
```

16. Print the following pattern. Get the number of lines to be printed from the user.

```
1
12
123
1234
```

17. Print the following pattern. Get the number of lines to be printed from the user.

```
1
12
123
1234
123
12
1
```